Onsective — Roadmap & System Plan

Onsective Engineering

2026-04-30

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Onsective — Roadmap & System Plan

Version: 1.0 **Date:** 2026-04-30 **Status:** Phased plan from current deployment to production-ready launch
Owner: Onsective engineering

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1. Executive Summary

Onsective is a multi-sided ecommerce marketplace. Sellers list products; buyers purchase; the platform takes a category-tiered 8 — 15% commission on every sale.

Three deployed surfaces:

Surface	Domain	Audience	Tech
Buyer marketplace	itsnottechy.cloud	Shoppers	Next.js (apps/web), port 3000
Seller portal	seller.itsnottechy.cloud	Sellers	Next.js (apps/seller), port 3002
Admin console	console.itsnottechy.cloud	Platform staff	Next.js (apps/console), port 3001

All three share one Postgres 16 database, one Redis instance (BullMQ + rate limiting), and one PM2 supervisor. Email is sent via Hostinger SMTP. Payments are processed via Stripe Connect Express — Onsective is the merchant of record; sellers are paid weekly via Stripe Transfers.

Where we are: auth (signup, email verify, 2FA, lockout, password reset, passkeys), the core schema (User, Seller, Product, Order, Payment, LedgerEntry, Payout), the buyer checkout flow, the seller application flow, and the admin console for users / sellers / orders / returns / tickets / approvals are all shipped and live.

Where we're going: a production-ready marketplace that takes real money, ships real packages, and has zero dead links or stub pages. This document defines the path in eleven phases.

Phase summary:

#	Phase	Effort	Blocking?
0	Stabilization	0.5 day	Yes (today)
1	Seller portal parity	1 day	Yes
2	Console gap-filling	1 day	Yes
3	Buyer-site completion	1.5 days	No
4	Catalog seeding + images	1 day	No
5	Production money infra	2-3 days	Yes (pre-launch)
6	Trust & safety v1	2 days	Yes (pre-launch)
7	Reviews + Q&A	3 days	No
8	Performance + scale	1 week	No
9	Launch readiness	1 week	Yes (pre-launch)
10	Growth surface	Ongoing	No

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2. System Architecture

2.1 Process & port topology

```
                    ┌──────────────────────────────────┐
                    │          Internet (HTTPS 443)    │
                    └──────────────────────────────────┘
                                   │
                                   ▼
                    ┌──────────────────────────────────┐
                    │     CloudPanel · Nginx (reverse  │
                    │     proxy + Let's Encrypt SSL)   │
                    └──────────────────────────────────┘
                                   │
                                   │
      ┌────────────────────┬────────────────────────┬────────────────────┐
      │                    │                        │                    │
      ▼                    ▼                        ▼                    │
itsnottechy.cloud   seller.itsnottechy.cloud   console.itsnottechy.cloud  │
      │                    │                        │                    │
      ▼                    ▼                        ▼                    │
apps/web (3000)     apps/seller (3002)        apps/console (3001)        │
                                                                         │
                                                                         ▼
                                                        PM2 (single supervisor)
                                                       ┌──────────────────────┐
                                                       │ web                  │
                                                       │ seller               │
                                                       │ console              │
                                                       │ worker-payouts       │
                                                       │ worker-images        │
                                                       │ worker-search-index  │
                                                       └──────────────────────┘
                                                                   │
                                                                   ▼
                                                       ┌──────────────────────┐
                                                       │ Postgres 16  (port 5432)│
                                                       │ Redis        (port 6379)│
                                                       └──────────────────────┘
```


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2.2 Data flow — buyer order

```
┌─────┐      1. POST /checkout/placeOrder      ┌──────────────────┐
│Buyer │──────────────────────────────────────►│  apps/web (tRPC)  │
└─────┘                                        └──────────┬───────┘
    ▲                                                     │
    │ 6. order placed (UI redirect)                       │ 2. INSERT Order, OrderItem
    │                                                     │    (commission frozen here)
    │                                                     ▼
    │                                          ┌──────────────────┐
    │                                          │ Postgres         │
    │                                          └──────────┬───────┘
    │                                                     │
    │  5. session continues                               │ 3. Stripe.paymentIntents
    │                                                     │       .create({on_behalf_of})
    │                                                     ▼
    │                                          ┌──────────────────┐
    │                                          │ Stripe API       │
    │                                          └──────────┬───────┘
    │                                                     │
    │ 4. payment_intent.succeeded webhook (signed)        │
    │ ────────────────────────────────────────────────────┘
    │                                                     ▼
    │                                          ┌──────────────────┐
    │                                          │ webhook handler  │
    │                                          │ - dedup via      │
    │                                          │   WebhookEvent   │
    │                                          │ - post LedgerEntry │
    │                                          │   (DR/CR balanced) │
    │                                          │ - decrement stock  │
    │                                          └──────────────────┘
```


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2.3 Money flow — seller payout

```
                          commission                  shipping
                          (8–15%)                     (carrier cost)
                             │                            │
buyer                        ▼                            ▼
  │   $100   ┌──────────┬────────┬──────────┬──────┬──────────┐
  │ ────────►│ Onsective│ –8 %   │ –shipping│ –fees│ = seller│  $87 (example)
  │   pay    │ collects │ comm   │ ($5)     │ ($3) │ payable │
  ▼          └──────────┴────────┴──────────┴──────┴──────────┘
Stripe                                                  │
                                                        │ held 7 days
                                                        │ (return window)
                                                        ▼
                                          ┌──────────────────────┐
                                          │ payouts worker (hourly)  │
                                          │  – groupBy seller        │
                                          │  – skip suspended        │
                                          │  – skip if Stripe        │
                                          │    payouts disabled      │
                                          │  – call                  │
                                          │    stripe.transfers      │
                                          │      .create()           │
                                          └──────────────────────┘
                                                     │
                                                     ▼
                                          ┌──────────────────────┐
                                          │ seller's Stripe Connect  │
                                          │   account                │
                                          └──────────────────────┘
                                                     │
                                                     ▼
                                          ┌──────────────────────┐
                                          │ seller's bank (~2–3 days)│
                                          └──────────────────────┘
```


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2.4 Auth flow — buyer signup with 2FA

```
┌─────┐            /signup form          ┌──────────┐
│Buyer │ ──────────────────────────────► │ apps/web  │
└─────┘                                  └──────────┘
                                                 │
                                                 │ 1. INSERT User
                                                 │    (passwordHash, emailVerified=null)
                                                 │ 2. INSERT Otp (purpose=verify)
                                                 │ 3. SMTP send 6-digit code
                                                 ▼
                                          ┌──────────┐
                                          │ Hostinger │
                                          │ SMTP      │
                                          └──────────┘
                                                 │
                                                 ▼
                                          ┌──────────┐
                                          │  inbox    │
                                          └──────────┘
                                                 │
   /verify-email form ◄──────────────────────────┘
   user enters code
                                                 │
                                                 ▼
                                          ┌──────────┐
                                          │ apps/web  │
                                          └──────────┘
                                                 │ 4. UPDATE User SET emailVerified=NOW()
                                                 │ 5. INSERT Session (httpOnly cookie)
                                                 │
   Subsequent login:
   /login (email + password)
                                                 │
                                                 │ 6. password match → INSERT Otp
(purpose=login_2fa)
                                                 │ 7. SMTP send 2FA code
                                                 ▼
   /verify-2fa form
   user enters code
                                                 │ 8. UPDATE User SET lastLogin/IP/UA
                                                 │ 9. INSERT Session
                                                 │ 10. detect new device → SMTP alert email
                                                 ▼
                                            buyer signed in
```

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3. Current-State Audit

Status legend: ✓ shipped · ◐ stub or partial · ✗ missing · ⚠ broken

3.1 Buyer marketplace (itsnottechy.cloud)

Area	Status	Notes
Signup / login / 2FA	✓	Working end-to-end
Password reset	✓	Working end-to-end
Email lockout (5 fails)	✓	Working
Passkeys (WebAuthn)	✓	Working; null/null challenge fix applied
Account hub / settings	✓	`/account` index plus subpages
Header + footer	✓	Color-blocked bento marketplace style
Product detail page	✓	Live with variants and gallery
Category pages	✓	Server-rendered, paginated
Cart	✓	Live tRPC backed
3-step checkout	✓	Address → shipping → pay (Stripe Elements)
Search (Postgres FTS)	✓	Falls back to ILIKE on small datasets
Sponsored ad slate	✓	On `/search`, second-price clearing
Order history	✓	`/account/orders` list + detail
Returns request flow	✓	Per-item from order detail
Wishlist link	⚠	Header link; `/account/wishlist` returns 404
Categories index	⚠	`/categories` link in header; page does not exist
Today's Deals	⚠	`/deals` link in header; page does not exist
Best Sellers	⚠	`/best-sellers` linked; page does not exist
Trending	⚠	`/trending` linked from home; page does not exist
Order tracking	✗	`/track` linked from footer; page does not exist
Address book CRUD	✗	Schema exists, no UI
Hindi translations	◐	Some keys missing
Mobile menu	◐	Desktop nav only — no hamburger sheet

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3.2 Seller portal (seller.itsnottechy.cloud)

Area	Status	Notes
Landing page	✓	Marketing page when logged out
Signup / apply form	✓	Form persists Seller row, status PENDING_KYC
Login	✓	Cookie-based, lockout enforced
Status dashboard	✓	4-step onboarding checklist
Edit storefront profile	✗	No UI exists
KYC document upload	✗	Apply form collects nothing; admin expects KYC docs
Stripe Connect onboarding	⚠	Button only on legacy `/[locale]/seller/*` (cross-domain)
Product list	⚠	Cross-domain link to itsnottechy.cloud — cookie not shared
Product create wizard (5-step)	⚠	Same — only on legacy app
Orders queue	⚠	Same
Payouts ledger	⚠	Same
Analytics dashboard	⚠	Same
tRPC client	✗	Currently uses raw fetch + REST routes

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3.3 Admin console (console.itsnottechy.cloud)

Area	Status	Notes
Login	✓	Email + password; PRIVILEGED_ROLES gate
IP allowlist	✓	Empty allowlist = open (dev)
Users list / detail	✓	Search, role editor, suspend, reactivate, delete
Create user	✓	Sends welcome email with set-password code
Sellers approval queue	✓	List + detail + approve/reject
KYC document review UI	✗	KycDocument schema exists; no UI
Orders queue / detail	✓	Cancel + refund (4-eyes if > $500)
Returns queue	◐	List + detail page exist; 4 state-machine action buttons missing on detail
Tickets workspace	✓	Reply, internal note, status transitions
Organizations (B2B)	✓	Approve, set credit limit, suspend
Approvals (4-eyes)	✓	Self-approval blocked
Ad moderation	✓	Approve drafts → ACTIVE
Audit log search UI	✗	AuditLog table exists; no page
Webhook event inspector	✗	WebhookEvent table exists; no page
System health page	✗	No `/dashboard/health`
Impersonation	✓	Magic-link, single-use, audited, read-only
SLA breach indicators	✗	Tickets have no breached/healthy badge

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3.4 Cross-cutting infrastructure

Area	Status	Notes
Postgres backups	✗	No automated `pg_dump`
Monitoring / Sentry	◐	Sentry SDK wired, DSN not set in production
Health check endpoint	✗	No `/api/health`
OpenSearch	✗	Code wired; cluster not provisioned
EasyPost live carrier rates	✗	Test API key; checkout uses hardcoded $5 shipping
Stripe live mode	✗	Test keys; placeholders in `.env`
S3 + CloudFront for images	✗	Bucket not provisioned
Image variants worker	◐	Code exists; never runs because no uploads happen
SES / SPF / DKIM / DMARC	⚠	Hostinger SMTP works; DNS records not set — emails risk spam folder
Status page	✗	None

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4. Core User Workflows

4.1 Buyer journey (target end-state)

```
   Discover → Browse → Compare → Cart → Sign in → Address → Ship → Pay → Track → Receive
→ Review
      |          |         |         |        |          |         |      |        |
|           |
      |          |         |         |        |          |         |      |        |
|           |
   /home       /category   PDP       /cart   /login     /checkout /shipping /pay   /account/  de-
livery        review
                                                        /address                            or-
ders/[n]    notification
```

Critical-path coverage: ✓ except review notification email (Phase 7), tracking page (Phase 3).

4.2 Seller journey (target end-state)

```
   Discover → Apply → KYC → Stripe → List → Sell → Ship → Get paid → Analyze
      |          |        |       |        |        |      |        |             |
      |          |        |       |        |        |      |        |             |
   seller.    /signup   /dashboard /dashboard /products /orders /orders/  /payouts   /analytics
   itsnot.   →/apply    /kyc      /connect   /new      queue   [n]/ship             ranges
   techy.cloud
```

Critical-path coverage: ◐ — Phase 1 closes the gaps (storefront edit, KYC upload, on-domain Stripe + product wizard + orders + payouts + analytics).

4.3 Admin journey (target end-state)

```
   Login → Triage → Approve → Moderate → Resolve → Audit
      |        |          |           |             |          |
      |        |          |           |             |          |
   /login   /dashboard   /sellers    /dashboard   /tickets/[n] /audit
                         /[id]      /ads          /returns/[n]
                         (KYC)      /products
                                   /approvals
```

Critical-path coverage: ✓ for users, sellers, orders, tickets, approvals; ✗ for KYC review (Phase 2), audit search (Phase 2), webhook inspection (Phase 2), system health (Phase 2).

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4.4 Money flow (lifecycle)

```
T=0     Buyer pays   ⟶ Onsective Stripe balance
T=0     payment_intent.succeeded ⟶ LedgerEntry posted (commission frozen at OrderItem)
T+0     OrderItem.status = PAID
T+1d    Seller ships ⟶ Shipment.status = IN_TRANSIT
T+5d    EasyPost webhook ⟶ Shipment.status = DELIVERED
T+7d    Return window closes ⟶ Payouts worker eligible
T+8d    Hourly cron runs ⟶ stripe.transfers.create() to seller
T+10d   Stripe ACH lands  ⟶ payout.paid webhook ⟶ Payout.status = PAID
```

Disputes: if `charge.dispute.created` fires before T+8d, the seller payable is held. After T+8d, the dispute is debited from the seller's next payout (this part needs hardening in Phase 5).

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5. Phased Roadmap

Each phase is sized to ship in one deploy. Phases are designed so a partial completion still leaves the system in a working state.

Phase 0 — Stabilization (today, ~30 minutes)

Goal: kill all current 404s, dead links, and routing bugs so the system feels coherent end-to-end.

Scope:

- Fix CloudPanel reverse-proxy URLs for all three sites
- Bootstrap admin user via SQL (`admin@onsective.com` → OWNER)
- Add SPF, DKIM, and DMARC TXT records in Hostinger DNS for `onsective.com`
- Confirm all 6 PM2 processes online + restart-on-boot via `pm2 startup`
- Smoke-test signup, 2FA, login, password reset on each surface

Out of scope: anything that requires code changes.

Dependencies: none.

Deliverables:

1. CloudPanel reverse-proxy URLs map cleanly: web→3000, console→3001, seller→3002
2. DNS records published and verified via `dig` / mxtoolbox
3. PM2 systemd unit installed; survives reboot
4. Admin user can log in to console

Success criteria: every documented URL returns 200 or 307; emails to Gmail land in inbox, not spam.

Phase 1 — Seller portal feature parity (1 deploy, ~6 hours)

Goal: `seller.itsnottechy.cloud` is fully self-contained. Zero cross-domain links. The legacy `/[locale]/seller/*` routes on the buyer app are deleted.

Scope:

1. Move into `apps/seller/dashboard`:
   - `/products` (list + 5-step wizard)
   - `/orders` (list + per-order detail)
   - `/payouts` (ledger view)
   - `/analytics` (charts + range picker)
   - `/onboarding/return` and `/onboarding/refresh` (Stripe Connect callback URLs)
2. Add `/dashboard/storefront` — edit form for legalName, displayName, description, country, tax id, logo
3. Add `/dashboard/kyc` — upload form for KycDocument with five doc types

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4. Add `/dashboard/connect` — Stripe Connect Express button
5. Build a tRPC client + provider for `apps/seller`
6. Delete `apps/web/src/app/[locale]/seller` and `apps/web/src/app/[locale]/sell`
7. Update Stripe Connect return / refresh URL constants to seller subdomain

Out of scope: any work in Phase 2+.

Dependencies: Phase 0.

Deliverables:

1. New seller signs up, fills profile, uploads KYC, gets approved, connects Stripe, lists a product, sees an order — all without leaving `seller.itsnottechy.cloud`
2. Buyer site has zero links to `/seller/*`
3. tRPC subset (`auth`, `seller`, `me`) wired to seller app

Success criteria: end-to-end seller flow works on the seller subdomain alone.

Phase 2 — Console gap-filling (1 deploy, ~5 hours)

Goal: admins can run the platform from the console without ever needing SQL.

Scope:

1. KYC review UI on `/dashboard/sellers/[id]` — list KycDocument rows, per-doc approve/reject buttons, bulk approve seller
2. Returns workflow buttons on `/dashboard/returns/[id]` — wire 4 server actions (approve, reject, mark-received, mark-refunded) that already exist
3. Audit log search at `/dashboard/audit` — search by actor, action, date range, target id; pagination
4. Webhook event inspector at `/dashboard/webhooks` — viewer, retry-failed, mark-resolved
5. System health at `/dashboard/health` — Postgres latency, Redis ping, BullMQ queue depths, worker last-heartbeat
6. Seller management actions — change commission %, force-disconnect Stripe, reactivate from suspended
7. Ticket SLA breach indicators

Dependencies: Phase 0.

Deliverables: all six pages above, all buttons wired and audited.

Success criteria: PM can run the platform for one week without SSH-ing in.

Phase 3 — Buyer site completion (1 deploy, ~6 hours)

Goal: every link on the buyer site goes somewhere real. Mobile menu works.

Scope:

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1. Build missing pages:
   - `/categories` — full grid
   - `/deals` — products with `mrpAmount > priceAmount`, sorted by % off
   - `/best-sellers` — order by `_count.orderItems`
   - `/trending` — last-24h order-frequency leaderboard
   - `/track/[orderNumber]` — guest-accessible (email-protected)
   - `/account/addresses` — CRUD on the existing `Address` model
   - `/account/wishlist` — Wishlist model + item add/remove
2. New `Wishlist` schema + tRPC endpoints
3. Search filters — price range, category, brand, seller rating
4. Mobile menu (hamburger sheet)
5. Hindi translations completed
6. Empty-state polish

Dependencies: Phase 0.

Deliverables: seven new pages, one new schema, complete translation file.

Success criteria: open the site on a phone, click every nav item, never hit 404.

Phase 4 — Catalog seeding + image pipeline (1 deploy, ~4 hours)

Goal: marketplace doesn't look dead. Real-feel browsing experience pre-launch.

Scope:

1. Provision real S3 bucket (`onsective-products`) + CloudFront distribution + OAC
2. Wire image variants worker to actually run (sharp pipeline coded; needs to enqueue on real upload)
3. Seed script with 4 demo seller accounts and 80 products distributed across 8 categories with Unsplash images
4. Realistic prices, descriptions, ratings (50-2000 reviews), variant data
5. Mark all demo data with `isDemo: true` flag — wipeable before launch
6. Demo sellers excluded from real Stripe transfers

Dependencies: AWS account; S3 credentials.

Deliverables: S3 + CloudFront live, seed script committed, 80 demo products visible.

Success criteria: load `itsnottechy.cloud/category/electronics` and see 10+ realistic products with images, ratings, prices.

Phase 5 — Production money infrastructure (2-3 days, blocked on external services)

Goal: ready to take real money from real buyers.

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Scope:

1. Stripe live mode — switch all `STRIPE_*_KEY` env vars from test to live; Stripe Connect Express live; Stripe Tax enabled; webhook endpoint registered with all events from DEPLOY.md §7
2. EasyPost real account — pay for an account, set live API key + webhook secret, register webhook
3. Email — choose between AWS SES (better deliverability) or stay on Hostinger with full SPF / DKIM / DMARC
4. TLS auto-renewal — confirm Let's Encrypt certbot cron is running
5. Postgres backups — `pg_dump` to S3 daily via cron + restore drill
6. Monitoring — Sentry DSN set in production env, UptimeRobot pinging the app
7. `/api/health` endpoint — returns 200 if Postgres, Redis, and queue workers are healthy
8. Rate limit hardening — apply `publicReadRateLimit` to all catalog reads
9. Live-key dispute handling — verify `charge.dispute.created` flow against real Stripe sandbox

Dependencies: Phase 0-4 complete; legal entity (Onsective LLC or similar) registered; bank account for Stripe payouts.

Deliverables: real $50 order placed by real buyer, shipped, delivered, paid out — fully hands-off.

Success criteria: ten test orders processed end-to-end; books balance; Stripe reconciliation report matches LedgerEntry totals.

Phase 6 — Trust & safety v1 (1-2 days)

Goal: can survive a determined troll and pass a basic security audit.

Scope:

1. 2FA recovery codes — 10 single-use codes shown after enabling 2FA
2. SMS OTP via Twilio — for users without email access
3. GDPR data export at `/account/privacy` — JSON dump emailed
4. Account deletion flow — proper hard-delete with PII purge after 30-day grace
5. Cookie consent banner for EU visitors (geolocate by countryCode)
6. Privacy policy + Terms of Service pages — boilerplate templates, real legal review before launch
7. Suspicious-login lockdown — 3 different IPs failing on same email in 1h = 24h lock
8. Reverse-proxy security headers — CSP, HSTS, frame-ancestors

Dependencies: Twilio account, real legal review.

Deliverables: all eight items shipped + signed legal docs.

Success criteria: external security review finds no P0; OWASP Top 10 checklist clean.

Phase 7 — Reviews, Q&A, ratings (2-3 days)

Goal: social proof on every product detail page.

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Scope:

1. Buyer can leave a review after order DELIVERED
2. Seller can respond to reviews
3. PDP shows aggregate rating, distribution chart, recent reviews
4. Verified-purchase badge
5. Q&A section on PDP — buyer asks, seller answers
6. Console moderation queue for review reports
7. Search results show rating star + count

Dependencies: Phase 4 (real catalog).

Deliverables: Review and Question / Answer schemas + UI; review-prompt email template.

Success criteria: buyer who placed three orders sees three review-prompt emails after delivery and can leave reviews.

Phase 8 — Performance + scale prep (1 week)

Goal: site stays fast as catalog grows from 80 to 80,000 products.

Scope:

1. OpenSearch turn-up — provision cluster, run `pnpm reindex:search`, switch from Postgres FTS
2. CDN — Cloudflare in front of `itsnottechy.cloud`
3. DB indexes audit — `EXPLAIN ANALYZE` 10 slowest queries
4. N+1 detection — Prisma `$on('query')` logging, fix worst offenders
5. Image lazy loading + AVIF — `next/image` everywhere
6. Bundle-size audit — `@next/bundle-analyzer`
7. Lighthouse CI — gate on PRs

Dependencies: Phases 4-7.

Deliverables: Lighthouse Performance > 90 on home + PDP; p99 server response < 300ms.

Success criteria: k6 load test passes 100 RPS sustained.

Phase 9 — Launch readiness (1 week)

Goal: ready for first 100 real customers.

Scope:

1. Load test — k6 with 100 concurrent users, find breaking point
2. Status page at `status.onsective.com` (UptimeRobot or BetterStack)
3. On-call rotation + PagerDuty integration with Sentry
4. Real customer support — `help@onsective.com` → Tickets workspace

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5. Documentation site at `docs.onsective.com`
6. Refund / return / shipping policy pages
7. First 10 sellers manually onboarded (white-glove)
8. Beta-launch announcement — Product Hunt, ads, social

Dependencies: Phases 5-8.

Deliverables: first real order placed, shipped, delivered, paid out without manual intervention.

Success criteria: 100 sellers onboarded in week one without on-call paging more than once per day.

Phase 10 — Growth surface (ongoing)

Goal: scale from 10 to 1000 sellers.

Scope: Onsective Prime subscription, sponsored ads UI + reporting, mobile apps (PWA + Capacitor), affiliate / referral program, marketing automation, B2B sales (NET_30 invoicing already schema'd).

Dependencies: Phase 9.

Deliverables: post-launch product roadmap.

Success criteria: retention curves stabilize; >40% of new sellers list a product within 7 days of approval.

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6. Risk Register

ID	Risk	Likelihood	Impact	Mitigation
R1	Email delivery (Gmail spam)	High	High	Phase 0 SPF/DKIM/DMARC; Phase 5 SES migration
R2	Stripe account suspension	Low	Critical	Maintain compliance; have backup processor (Adyen) plan
R3	Postgres data loss	Low	Critical	Phase 5 daily backups + restore drill
R4	Seller fraud (stolen cards used to buy own products)	Medium	Medium	Stripe Radar; manual review on first 10 orders/seller
R5	Buyer chargeback abuse	Medium	Medium	7-day return-window hold; require photo evidence on returns
R6	DDoS	Low	High	Phase 8 Cloudflare; rate limits already wired
R7	Tax compliance (GST/VAT/sales)	Medium	High	Phase 5 Stripe Tax; consult tax counsel pre-launch
R8	Image hosting costs balloon	Low	Medium	CloudFront + S3 lifecycle policy (Phase 8)
R9	OpenSearch cluster outage	Medium	Medium	Postgres FTS fallback already coded
R10	Founder bus factor	Always	Critical	Documentation; runbook; CI/CD
R11	EU GDPR fine	Low	High	Phase 6 data export + deletion + cookie consent
R12	Stripe payout reversal mid-flight	Low	High	Idempotency keys on every transfer (already wired)

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7. Appendices

Appendix A — Environment variable reference

Variable	Required	Purpose	Example
`DATABASE_URL`	Yes	Postgres connection	`postgresql://onsective:****@127.0.0.1:5432/onsective`
`REDIS_URL`	Yes	BullMQ + rate limit	`redis://127.0.0.1:6379`
`SESSION_SECRET`	Yes	Cookie signing	32-byte hex
`STRIPE_SECRET_KEY`	Yes	Stripe API	`sk_live_...`
`STRIPE_WEBHOOK_SECRET`	Yes	Webhook signature	`whsec_...`
`STRIPE_PUBLISHABLE_KEY`	Yes	Stripe Elements client	`pk_live_...`
`WEBAUTHN_RP_ID`	Yes	Passkey domain	`itsnottechy.cloud`
`WEBAUTHN_ORIGIN`	Yes	Passkey origin	`https://itsnottechy.cloud`
`SMTP_HOST` / `SMTP_USER` / `SMTP_PASS`	Yes	Outbound email	`smtp.hostinger.com`
`EMAIL_FROM`	Yes	Sender address	`donotreply@onsective.com`
`S3_BUCKET_PRODUCTS`	Phase 4+	Image storage	`onsective-products`
`EASYPOST_API_KEY`	Phase 5+	Carrier rates	`EZAK...`
`OPENSEARCH_URL`	Phase 8+	Search cluster	`https://search.onsective.com`
`SENTRY_DSN`	Phase 5+	Error tracking	`https://...@sentry.io/...`
`CONSOLE_IP_ALLOWLIST`	Phase 6+	Console access	`1.2.3.4,5.6.7.8`

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Appendix B — Critical tRPC routers

Router	Procedures	Purpose
auth	signup, login, verifyTwoFactor, requestPasswordReset, resetPassword, logout, passkeys.*	Account lifecycle
me	profile, addresses	Buyer self-service
catalog	categories.tree, products.byCategory, products.bySlug, search	Browse
cart	get, addItem, updateQty, removeItem	Cart state
checkout	summary, placeOrder	Checkout
order	byNumber, list	Buyer orders
seller	application.apply / status, dashboard.summary, products.*, orders.*, connect.*, uploads.*, analytics.*	Seller surface
admin	sellers.approve / reject, products.approve	Console actions
organizations	my, create, members, invite, removeMember, currentExemption	B2B
prime	status, startCheckout, cancel	Membership
ads	campaigns.*, buildSlate, trackImpression, trackClick	Sponsored ads
returns	list, request, cancel	Returns

Appendix C — Deployment runbook (current)

ONE-TIME BOOTSTRAP

1. **As root:** `apt install postgresql-16 redis-server libvips-dev`; install Node 20, pnpm 9, pm2
2. **As root:** `sudo -u postgres psql -c "CREATE USER onsective WITH PASSWORD '...';"`; `CREATE DATABASE onsective OWNER onsective;`
3. **CloudPanel UI:** create three Reverse Proxy sites (web→3000, console→3001, seller→3002)
4. **DNS:** A records for apex, console, seller subdomains
5. **As site user:** `git clone`, write `.env`, symlink into each app + db package, `pnpm install`, `pnpm db:generate`, `pnpm db:deploy`, `pnpm db:seed`, `pnpm build`
6. **As site user:** `pm2 start ecosystem.config.js`, `pm2 save`
7. **As root:** `pm2 startup systemd -u <site-user> --hp /home/<site-user>`
8. **CloudPanel UI:** SSL/TLS → Let's Encrypt for each site
9. **SQL:** promote one user to OWNER

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PER-DEPLOY

```
git pull origin main
set -a; source .env; set +a
pnpm install
pnpm db:generate
pnpm --filter @onsective/db run deploy
pnpm build
pm2 restart all --update-env
```

Appendix D — Schema highlights

User (with Phase 0+ security): `id, email, emailVerified, passwordHash, fullName, locale, countryCode, roles[], status, twoFactorEmail, failedLoginAttempts, lockedUntil, lastLoginIp, lastLoginUserAgent, lastLoginAt, deletedAt, ...`

Seller: `id, userId, legalName, displayName, slug, description, countryCode, taxId, status (PENDING_KYC|KYC_SUBMITTED|APPROVED|SUSPENDED|REJECTED), stripeAccountId, stripePayoutsEnabled, defaultCommissionPct, allowedDestinationCountries[], approvedAt, approvedBy, ...`

Product: `id, sellerId, categoryId, slug, title, description, brand, status (DRAFT|PENDING_REVIEW|ACTIVE|REJECTED|ARCHIVED), images[], variants[], hsCode, originCountryCode, ...`

Order: `id, orderNumber, buyerId, status, totals (sub/tax/shipping/discount/grand), currency, organizationId?, b2bInvoiceId?, taxExempt, ...`

OrderItem: `id, orderId, sellerId, productId, variantId, qty, priceAmount, mrpAmount, commissionPctBps (frozen here), status (CREATED|PAID|CONFIRMED|SHIPPED|DELIVERED|RETURNED|REFUNDED|CANCELLED), ...`

LedgerEntry: `id, postedAt, accountType (MERCHANT_RECEIVABLE|PLATFORM_REVENUE|SELLER_PAYABLE|SELLER_PAID|GATEWAY_FEE|REFUND|...), debitMinor, creditMinor, currency, sellerId?, orderItemId?, refundId?, payoutId?, paymentId?, metadata` — every set must balance (sum DR = sum CR).

Payout: `id, sellerId, periodStart, periodEnd, amountMinor, currency, status (PENDING|IN_TRANSIT|PAID|FAILED), stripeTransferId, ledgerEntries[]`

KycDocument: `id, sellerId, type (GOVT_ID|TAX_CERTIFICATE|BANK_STATEMENT|BUSINESS_REGISTRATION|ADDRESS_PROOF), status (PENDING|APPROVED|REJECTED), fileKey, ...`

Appendix E — Quick reference

Surfaces and ports

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Surface	Domain	Port	App
Buyer	itsnottechy.cloud	3000	apps/web
Console	console.itsnottechy.cloud	3001	apps/console
Seller	seller.itsnottechy.cloud	3002	apps/seller

PM2 processes (6 total)

`web`, `console`, `seller`, `worker-payouts`, `worker-images`, `worker-search-index`

Cron schedule

Job	Frequency	Command
Payouts sweep	Hourly UTC	enqueued via BullMQ repeatable
Ads daily reset	Hourly UTC	`pnpm cron:ads-reset`
Search reindex	Manual	`pnpm reindex:search`

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End of document

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